

Experiment No: 3 Exploratory Data Analysis(Data cleaning and data Transformation)

Aim:

To import and export different types of files using python.

Algorithm:

1. **Start** the program.
2. **Import** the required libraries (pandas, numpy).
3. **Load** the dataset into a DataFrame.
4. **Check** the dataset structure using info().
5. **Identify** missing values using isnull().sum().
6. **Handle** missing values by removing or replacing them.
7. **Check** for duplicate records using duplicated().
8. **Remove** duplicate records using drop_duplicates().
9. **Identify** outliers using statistical methods or box plots.
10. **Handle** outliers by removing or capping extreme values.
11. **Verify** that the cleaned dataset has no unwanted missing values or duplicates.
12. **Save** the cleaned dataset (optional).
13. **Stop** the program.

Program:

1.

```
""" 1.0 """

import pandas as pd

# Read CSV file
df = pd.read_csv(r"C:\Users\AI_LAB\Downloads\social_media_dataset_with_null_duplicates_outliers.

print(df.head())
print(df.tail())
print(df.info())
print(df.shape)
print(df.columns)
print(df.describe())
```

Output:

```
   Post_ID  Platform  Likes  Comments  Shares  Views  Sentiment
0         1  Instagram   304.0     384.0      71   17049  Positive
1         2   Facebook   939.0     351.0     190   36741  Positive
2         3   YouTube  250000.0       21.0       8    7140  Positive
3         4   Facebook  4239.0     313.0       7   37781  Positive
4         5   YouTube  3536.0     117.0     115   39618   Neutral
   Post_ID  Platform  Likes  Comments  Shares  Views  Sentiment
48        49   YouTube   220.0       52.0     193   16491  Positive
49        50  LinkedIn  4078.0     251.0      55   27282  Positive
50         9   Twitter   455.0     378.0     118   36142  Positive
51        16  LinkedIn  3208.0     143.0     164   46099  Negative
52        26  Instagram  4498.0     389.0      69   43006   Neutral
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 53 entries, 0 to 52
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Post_ID     53 non-null     int64
1   Platform    52 non-null     object
2   Likes       52 non-null     float64
3   Comments    52 non-null     float64
4   Shares       53 non-null     int64
5   Views       53 non-null     int64
6   Sentiment   52 non-null     object
dtypes: float64(2), int64(3), object(2)
memory usage: 3.0+ KB
None
(53, 7)
Index(['Post_ID', 'Platform', 'Likes', 'Comments', 'Shares', 'Views',
      'Sentiment'],
      dtype='object')
   Post_ID  Likes  Comments  Shares  Views
count  53.000000   52.000000  52.000000  53.000000  53.000000
mean    25.018868  7544.961538  345.115385   96.584906  45828.415094
std     14.386814 34306.916476  674.785214   61.786367 134280.320211
min      1.000000   220.000000    8.000000    5.000000   4665.000000
25%     13.000000  1893.750000  116.000000   51.000000  16491.000000
50%     25.000000  3038.500000  276.500000   76.000000  31318.000000
```


3.sum:

```
in [20]:  
print(df.isnull().sum())
```

Output:

```
Post_ID      0  
Platform      1  
Likes        1  
Comments     1  
Shares       0  
Views        0  
Sentiment    1  
dtype: int64
```

4 fillna:

```
df["Platform"] = df["Platform"].fillna(df["Platform"].mode()[0])  
  
df["Sentiment"] = df["Sentiment"].fillna(df["Sentiment"].mode()[0])  
  
print(df)
```

Output:

	Post_ID	Platform	Likes	Comments	Shares	Views	Sentiment
0	1	Instagram	304.0	384.0	71	17049	Positive
1	2	Facebook	939.0	351.0	190	36741	Positive
2	3	YouTube	250000.0	21.0	8	7140	Positive
3	4	Facebook	4239.0	313.0	7	37781	Positive
4	5	YouTube	3536.0	117.0	115	39618	Neutral
5	6	Instagram	NaN	362.0	109	23298	Neutral
6	7	Facebook	1863.0	495.0	196	23059	Positive
7	8	Instagram	3212.0	54.0	92	23541	Negative
8	9	Twitter	455.0	378.0	118	36142	Positive
9	10	LinkedIn	745.0	287.0	76	42198	Negative
10	11	Twitter	4829.0	103.0	181	5558	Positive
11	12	Facebook	2470.0	45.0	60	7619	Neutral
12	13	Instagram	3814.0	330.0	94	11659	Neutral
13	14	Twitter	1816.0	348.0	69	46994	Negative
14	15	Instagram	1501.0	278.0	187	17043	Positive
15	16	LinkedIn	3208.0	143.0	164	46099	Negative
16	17	Facebook	2756.0	436.0	197	4665	Positive
17	18	Instagram	2684.0	210.0	69	5337	Positive
18	19	YouTube	2677.0	113.0	168	1000000	Neutral
19	20	LinkedIn	1270.0	140.0	36	17162	Negative
20	21	YouTube	4515.0	NaN	192	39311	Neutral
21	22	YouTube	3371.0	190.0	57	10065	Negative
22	23	LinkedIn	844.0	391.0	13	8185	Positive
23	24	Facebook	3558.0	310.0	17	26216	Neutral
24	25	YouTube	3934.0	275.0	65	37256	Positive
25	26	Instagram	4498.0	389.0	69	43006	Neutral
26	27	Instagram	2504.0	227.0	41	30735	Positive
27	28	Twitter	4200.0	395.0	46	34271	Positive
28	29	Twitter	4258.0	316.0	51	11016	Neutral
29	30	Facebook	4518.0	493.0	200	35757	Positive
30	31	YouTube	2755.0	255.0	5	8331	Neutral
31	32	Twitter	2061.0	34.0	62	38182	Positive
32	33	Instagram	4081.0	422.0	18	35911	Positive
33	34	Facebook	3993.0	489.0	141	11821	Positive
34	35	YouTube	3566.0	498.0	55	36343	Negative
35	36	Facebook	2653.0	209.0	172	43590	Neutral
36	37	LinkedIn	4339.0	236.0	31	17246	Positive
37	38	Instagram	2869.0	15.0	151	37301	Positive

5 duplicate:

```
print(df.duplicated())
df.drop_duplicates(inplace=True)
print(df)
```

Output:

```
23    False
24    False
25    False
26    False
27    False
28    False
29    False
30    False
31    False
32    False
33    False
34    False
35    False
36    False
37    False
38    False
39    False
40    False
41    False
42    False
43    False
44    False
45    False
46    False
47    False
48    False
49    False
50     True
51     True
52     True
```

6 Outliers:

```
Q1 = df['Likes'].quantile(0.25)
Q3 = df['Likes'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df['Likes'] < lower) | (df['Likes'] > upper)]

print(outliers)
```

Output:

Post_ID	Platform	Likes	Comments	Shares	Views	Sentiment
3	YouTube	250000.0	21.0	8	7140	Positive

7 Transform:

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df[['Likes']] = scaler.fit_transform(df[['Likes']])

print(df[['Likes']])
```

Output:

```
22 0.002498
23 0.013364
24 0.014869
25 0.017127
26 0.009144
27 0.015934
28 0.016166
29 0.017207
30 0.010149
31 0.007370
32 0.015458
33 0.015105
34 0.013396
35 0.009741
36 0.016491
37 0.010605
38 0.006742
39 0.007026
40 0.007322
41 0.003855
42 0.015029
43 0.013656
44 0.002746
45 0.007671
46 0.013356
47 0.014052
48 0.000000
49 0.015446
```

8 Encoder:

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df["Platform"] = le.fit_transform(df["Platform"])
print(df)
```

Output:

	Post_ID	Platform	Likes	Comments	Shares	Views	Sentiment
0	1	1	0.000336	384.0	71	17049	2
1	2	0	0.002879	351.0	190	36741	2
2	3	4	1.000000	21.0	8	7140	2
3	4	0	0.016090	313.0	7	37781	2
4	5	4	0.013276	117.0	115	39618	1
5	6	1	NaN	362.0	109	23298	1
6	7	0	0.006578	495.0	196	23059	2
7	8	1	0.011979	54.0	92	23541	0
8	9	3	0.000941	378.0	118	36142	2
9	10	2	0.002102	287.0	76	42198	0
10	11	3	0.018452	103.0	181	5558	2
11	12	0	0.009008	45.0	60	7619	1
12	13	1	0.014389	330.0	94	11659	1
13	14	3	0.006390	348.0	69	46994	0
14	15	1	0.005129	278.0	187	17043	2
15	16	2	0.011963	143.0	164	46099	0
16	17	0	0.010153	436.0	197	4665	2
17	18	1	0.009865	210.0	69	5337	2
18	19	4	0.009837	113.0	168	1000000	1
19	20	2	0.004204	140.0	36	17162	0
20	21	4	0.017195	NaN	192	39311	1
21	22	4	0.012615	190.0	57	10065	0
22	23	2	0.002498	391.0	13	8185	2
23	24	0	0.013364	310.0	17	26216	1
24	25	4	0.014869	275.0	65	37256	2
25	26	1	0.017127	389.0	69	43006	1
26	27	1	0.009144	227.0	41	30735	2
27	28	3	0.015934	395.0	46	34271	2
28	29	3	0.016166	316.0	51	11016	1

9.dummies:

```
pd.get_dummies(df['Platform'])
```

Output:

	0	1	2	3	4
1	True	False	False	False	False
2	False	False	False	False	True
3	True	False	False	False	False
4	False	False	False	False	True
5	False	True	False	False	False
6	True	False	False	False	False
7	False	True	False	False	False
8	False	False	False	True	False
9	False	False	True	False	False
10	False	False	False	True	False
11	True	False	False	False	False
12	False	True	False	False	False
13	False	False	False	True	False
14	False	True	False	False	False
15	False	False	True	False	False
16	True	False	False	False	False
17	False	True	False	False	False
18	False	False	False	False	True
19	False	False	True	False	False
20	False	False	False	False	True